

# **BABU BANARASI DAS**

# **UNIVERSITY**

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## **SCHOOL OF COMPUTER APPLICATIONS**

### **No SQL Project On**

### **Library Management System Using MongoDB**

**Submitted to:**

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# ACKNOWLEDGEMENT

I sincerely express my gratitude to my teacher Mr. Harendra Singh for giving me the opportunity to work on this NoSQL project titled “Library management system Using MongoDB”.

This project helped me understand the basic concepts of NoSQL databases and learn how MongoDB is used to create, retrieve, update, Sorting and delete data. I also learned how different query operators such as greater than, less than, equal to, AND, and OR are used in MongoDB.

I am thankful to my teacher for the guidance and support provided throughout the completion of this project.

# INTRODUCTION

MongoDB is a popular NoSQL database management system that stores data in the form of documents. It is flexible, scalable, and easy to use for managing large amounts of data.

The Library Management System using MongoDB is designed to manage library book records efficiently. In this project, information about books such as Book ID, Book Name, Author, Price, Category, and Quantity is stored in a MongoDB collection.

This project demonstrates various MongoDB operations such as creating a database and collection, inserting records, displaying data, updating records, deleting records, filtering data using comparison operators, sorting, limiting results, searching by category, and counting documents.

The main purpose of this project is to understand the basic concepts and practical use of MongoDB and NoSQL databases through a simple Library Management System.

## OBJECTIVES

- To understand the basic concepts of MongoDB and NoSQL databases.
- To create and manage a database and collection in MongoDB.

- To insert and store library book records in a MongoDB collection.
- To perform basic CRUD operations such as insert, update, and delete.
- To use comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte.
- To perform data filtering using \$and and \$or operators.
- To sort, search, limit, and count book records in MongoDB.
- To understand the practical implementation of a Library Management System using MongoDB..

## 1. CREATE DATABASE

### MongoDB Query:

*use library*

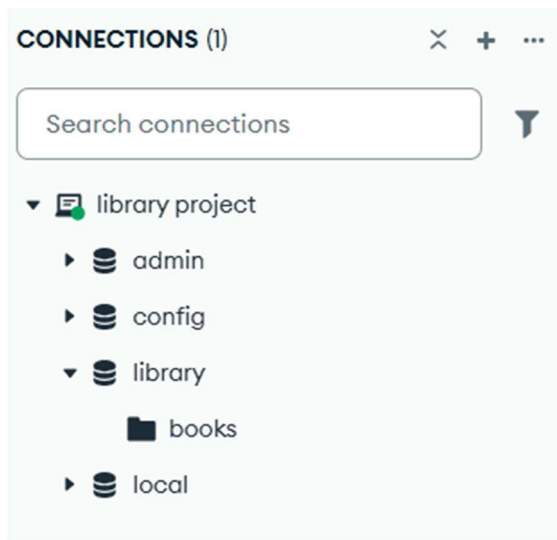
MongoDB uses the `use` command to create or switch to a database.

```
>_MONGOSH
> use("library")
< switched to db library
```

## 2. CREATE COLLECTION

### MongoDB Query:

`db.createCollection("books")`



## 3. INSERT 50 STUDENTS

### MongoDB Query:

```
db.books.insertMany([

{bookId:1,bookName:"Python Basics",author:"Mark
Lutz",price:500,category:"Technology",quantity:10},

{bookId:2,bookName:"Java Programming",author:"Herbert
Schildt",price:700,category:"Technology",quantity:8},

{bookId:3,bookName:"Data Science",author:"John
Smith",price:800,category:"Technology",quantity:6},

{bookId:4,bookName:"Machine Learning",author:"Tom
Mitchell",price:900,category:"Technology",quantity:5},

{bookId:5,bookName:"Atomic Habits",author:"James Clear",price:500,category:"Self
Help",quantity:12},

{bookId:6,bookName:"The Alchemist",author:"Paulo
Coelho",price:300,category:"Fiction",quantity:7},

{bookId:7,bookName:"Rich Dad Poor Dad",author:"Robert
Kiyosaki",price:450,category:"Finance",quantity:9},

{bookId:8,bookName:"Ikigai",author:"Hector Garcia",price:350,category:"Self
Help",quantity:11},

{bookId:9,bookName:"Harry Potter",author:"J.K.
Rowling",price:600,category:"Fantasy",quantity:6},

{bookId:10,bookName:"Wings of Fire",author:"A.P.J. Abdul
Kalam",price:250,category:"Biography",quantity:10},

{bookId:11,bookName:"Clean Code",author:"Robert
Martin",price:850,category:"Technology",quantity:5},

{bookId:12,bookName:"C Programming",author:"Dennis
Ritchie",price:400,category:"Technology",quantity:8},

{bookId:13,bookName:"Think and Grow Rich",author:"Napoleon
Hill",price:300,category:"Self Help",quantity:9},
```

**{bookId:14,bookName:"The Hobbit",author:"J.R.R. Tolkien",price:550,category:"Fantasy",quantity:7},**

**{bookId:15,bookName:"1984",author:"George Orwell",price:450,category:"Fiction",quantity:6},**

**{bookId:16,bookName:"SQL Fundamentals",author:"John Watson",price:650,category:"Technology",quantity:8},**

**{bookId:17,bookName:"Deep Work",author:"Cal Newport",price:500,category:"Self Help",quantity:10},**

**{bookId:18,bookName:"Atomic Structure",author:"R. Sharma",price:200,category:"Education",quantity:12},**

**{bookId:19,bookName:"Digital Marketing",author:"Philip Kotler",price:750,category:"Business",quantity:5},**

**{bookId:20,bookName:"Business Studies",author:"Poonam Gandhi",price:350,category:"Business",quantity:8},**

**{bookId:21,bookName:"Operating Systems",author:"Galvin",price:800,category:"Technology",quantity:6},**

**{bookId:22,bookName:"Computer Networks",author:"Andrew Tanenbaum",price:850,category:"Technology",quantity:5},**

**{bookId:23,bookName:"Database Systems",author:"Raghu Ramakrishnan",price:900,category:"Technology",quantity:4},**

**{bookId:24,bookName:"Artificial Intelligence",author:"Stuart Russell",price:950,category:"Technology",quantity:5},**

**{bookId:25,bookName:"The Psychology of Money",author:"Morgan Housel",price:600,category:"Finance",quantity:9},**

**{bookId:26,bookName:"Rich Dad Poor Dad 2",author:"Robert Kiyosaki",price:450,category:"Finance",quantity:7},**

**{bookId:27,bookName:"Sherlock Holmes",author:"Arthur Conan Doyle",price:300,category:"Fiction",quantity:6},**

**{bookId:28,bookName:"The Kite Runner",author:"Khaled Hosseini",price:400,category:"Fiction",quantity:5},**

**{bookId:29,bookName:"Harry Potter 2",author:"J.K. Rowling",price:650,category:"Fantasy",quantity:8},**

**{bookId:30,bookName:"The Power of Now",author:"Eckhart Tolle",price:550,category:"Self Help",quantity:10},**

**{bookId:31,bookName:"JavaScript Guide",author:"David Flanagan",price:700,category:"Technology",quantity:5},**

**{bookId:32,bookName:"Web Development",author:"Jon Duckett",price:800,category:"Technology",quantity:6},**

**{bookId:33,bookName:"HTML and CSS",author:"Jon Duckett",price:600,category:"Technology",quantity:8},**

**{bookId:34,bookName:"Python Crash Course",author:"Eric Matthes",price:750,category:"Technology",quantity:7},**

**{bookId:35,bookName:"Data Analytics",author:"James Evans",price:650,category:"Technology",quantity:6},**

**{bookId:36,bookName:"Statistics Basics",author:"S.C. Gupta",price:250,category:"Education",quantity:10},**

**{bookId:37,bookName:"Physics Fundamentals",author:"H.C. Verma",price:500,category:"Education",quantity:8},**

**{bookId:38,bookName:"Chemistry Guide",author:"O.P. Tandon",price:450,category:"Education",quantity:7},**

**{bookId:39,bookName:"Mathematics",author:"R.D. Sharma",price:550,category:"Education",quantity:9},**

**{bookId:40,bookName:"The Secret",author:"Rhonda Byrne",price:400,category:"Self Help",quantity:10},**

**{bookId:41,bookName:"Think Like a Monk",author:"Jay Shetty",price:600,category:"Self Help",quantity:8},**

```
{bookId:42,bookName:"The Fault in Our Stars",author:"John  
Green",price:350,category:"Fiction",quantity:6},  
  
{bookId:43,bookName:"Pride and Prejudice",author:"Jane  
Austen",price:450,category:"Fiction",quantity:5},  
  
{bookId:44,bookName:"The Lord of the Rings",author:"J.R.R.  
Tolkien",price:900,category:"Fantasy",quantity:4},  
  
{bookId:45,bookName:"The Chronicles of Narnia",author:"C.S.  
Lewis",price:700,category:"Fantasy",quantity:6},  
  
{bookId:46,bookName:"Startup Ideas",author:"Peter  
Thiel",price:800,category:"Business",quantity:5},  
  
{bookId:47,bookName:"Marketing Management",author:"Philip  
Kotler",price:850,category:"Business",quantity:6},  
  
{bookId:48,bookName:"Economics Today",author:"Paul  
Samuelson",price:500,category:"Business",quantity:7},  
  
{bookId:49,bookName:"The Lean Startup",author:"Eric  
Ries",price:750,category:"Business",quantity:8},  
  
{bookId:50,bookName:"Zero to One",author:"Peter  
Thiel",price:650,category:"Business",quantity:6}  
])
```

```

db.books.insertMany([
  {bookId:2,bookName:"Java Programming",author:"Herbert Schildt",price:700,category:"Technology",quantity:8},
  {bookId:3,bookName:"Data Science",author:"John Smith",price:800,category:"Technology",quantity:6},
  {bookId:4,bookName:"Machine Learning",author:"Tom Mitchell",price:900,category:"Technology",quantity:5},
  {bookId:5,bookName:"Atomic Habits",author:"James Clear",price:500,category:"Self Help",quantity:12},
  {bookId:6,bookName:"The Alchemist",author:"Paulo Coelho",price:300,category:"Fiction",quantity:7},
  {bookId:7,bookName:"Rich Dad Poor Dad",author:"Robert Kiyosaki",price:450,category:"Finance",quantity:9},
  {bookId:8,bookName:"Ikigai",author:"Hector Garcia",price:350,category:"Self Help",quantity:11},
  {bookId:9,bookName:"Harry Potter",author:"J.K. Rowling",price:600,category:"Fantasy",quantity:6},
  {bookId:10,bookName:"Wings of Fire",author:"A.P.J. Abdul Kalam",price:250,category:"Biography",quantity:10},
  {bookId:11,bookName:"Clean Code",author:"Robert Martin",price:850,category:"Technology",quantity:5},
  {bookId:12,bookName:"C Programming",author:"Dennis Ritchie",price:400,category:"Technology",quantity:8},
  {bookId:13,bookName:"Think and Grow Rich",author:"Napoleon Hill",price:300,category:"Self Help",quantity:9},
  {bookId:14,bookName:"The Hobbit",author:"J.R.R. Tolkien",price:550,category:"Fantasy",quantity:7},
  {bookId:15,bookName:"1984",author:"George Orwell",price:450,category:"Fiction",quantity:6},
  {bookId:16,bookName:"SQL Fundamentals",author:"John Watson",price:650,category:"Technology",quantity:8},
  {bookId:17,bookName:"Deep Work",author:"Cal Newport",price:500,category:"Self Help",quantity:10},
  {bookId:18,bookName:"Atomic Structure",author:"R. Sharma",price:200,category:"Education",quantity:12},
  {bookId:19,bookName:"Digital Marketing",author:"Philip Kotler",price:750,category:"Business",quantity:5},
  {bookId:20,bookName:"Business Studies",author:"Poonam Gandhi",price:350,category:"Business",quantity:8},
  {bookId:21,bookName:"Operating Systems",author:"Galvin",price:800,category:"Technology",quantity:6},
  {bookId:22,bookName:"Computer Networks",author:"Andrew Tanenbaum",price:850,category:"Technology",quantity:5},
  {bookId:23,bookName:"Database Systems",author:"Raghu Ramakrishnan",price:900,category:"Technology",quantity:4},
  {bookId:24,bookName:"Artificial Intelligence",author:"Stuart Russell",price:950,category:"Technology",quantity:5},
  {bookId:25,bookName:"The Psychology of Money",author:"Morgan Housel",price:600,category:"Finance",quantity:9},
  {bookId:26,bookName:"Rich Dad Poor Dad 2",author:"Robert Kiyosaki",price:450,category:"Finance",quantity:7},
  {bookId:27,bookName:"Sherlock Holmes",author:"Arthur Conan Doyle",price:300,category:"Fiction",quantity:6},
  {bookId:28,bookName:"The Kite Runner",author:"Khaled Hosseini",price:400,category:"Fiction",quantity:5}
])

```

## 4. DISPLAY ALL STUDENTS

MongoDB Query:

```
db.books.find()
```

Explanation: Retrieves documents from the books collection.

```
> db.books.find()
{
  _id: ObjectId('6ab295f279f427617fd6af9f'),
  bookId: 1,
  bookName: 'Python Basics',
  author: 'Mark Lutz',
  price: 500,
  category: 'Technology',
  quantity: 10
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa0'),
  bookId: 2,
  bookName: 'Java Programming',
  author: 'Herbert Schildt',
  price: 700,
  category: 'Technology',
  quantity: 8
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa1'),
  bookId: 3,
  bookName: 'Data Science',
  author: 'John Smith',
  price: 800,
  category: 'Technology',
  quantity: 6
}
```

## 5. EQUAL TO — \$eq

### MongoDB Query:

```
db.books.find({price: {$eq: 500}})
```

Explanation: Finds books whose price is equal to 500.

```
> db.books.find({price: {$eq: 500}})
< {
  _id: ObjectId('6ab295f279f427617fd6af9f'),
  bookId: 1,
  bookName: 'Python Basics',
  author: 'Mark Lutz',
  price: 500,
  category: 'Technology',
  quantity: 10
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa3'),
  bookId: 5,
  bookName: 'Atomic Habits',
  author: 'James Clear',
  price: 500,
  category: 'Self Help',
  quantity: 12
}
{
  _id: ObjectId('6ab2971a79f427617fd6afaf'),
  bookId: 17,
  bookName: 'Deep Work',
  author: 'Cal Newport',
  price: 500,
  category: 'Self Help',
  quantity: 10
}
```

## 6. GREATER THAN — \$gt

MongoDB Query:

```
db.books.find({price: {$gt: 700}})
```

Explanation: Finds books whose price is greater than 700.

```
> db.books.find({price: {$gt: 700}})
< {
  _id: ObjectId('6ab2971a79f427617fd6afa1'),
  bookId: 3,
  bookName: 'Data Science',
  author: 'John Smith',
  price: 800,
  category: 'Technology',
  quantity: 6
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa2'),
  bookId: 4,
  bookName: 'Machine Learning',
  author: 'Tom Mitchell',
  price: 900,
  category: 'Technology',
  quantity: 5
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa9'),
  bookId: 11,
  bookName: 'Clean Code',
  author: 'Robert Martin',
  price: 850,
  category: 'Technology',
  quantity: 5
}
```

## 7. LESS THAN — \$lt

MongoDB Query:

```
db.books.find({price: {$lt: 300}})
```

Explanation: Finds books whose price is less than 300.

```
> db.books.find({price: {$lt: 300}})
< {
  _id: ObjectId('6ab2971a79f427617fd6afa8'),
  bookId: 10,
  bookName: 'Wings of Fire',
  author: 'A.P.J. Abdul Kalam',
  price: 250,
  category: 'Biography',
  quantity: 10
}
{
  _id: ObjectId('6ab2971a79f427617fd6afb0'),
  bookId: 18,
  bookName: 'Atomic Structure',
  author: 'R. Sharma',
  price: 200,
  category: 'Education',
  quantity: 12
}
{
  _id: ObjectId('6ab2971a79f427617fd6afc2'),
  bookId: 36,
  bookName: 'Statistics Basics',
  author: 'S.C. Gupta',
  price: 250,
  category: 'Education',
  quantity: 10
}
```

## 8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

```
db.books.find({price: {$gte: 800}})
```

Explanation: Finds books whose price is greater than or equal to 800.

```
> db.books.find({price: {$gte: 800}})
< {
  _id: ObjectId('6ab2971a79f427617fd6afa1'),
  bookId: 3,
  bookName: 'Data Science',
  author: 'John Smith',
  price: 800,
  category: 'Technology',
  quantity: 6
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa2'),
  bookId: 4,
  bookName: 'Machine Learning',
  author: 'Tom Mitchell',
  price: 900,
  category: 'Technology',
  quantity: 5
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa9'),
  bookId: 11,
  bookName: 'Clean Code',
  author: 'Robert Martin',
  price: 850,
  category: 'Technology',
  quantity: 5
}
```

## 9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query:

```
db.books.find({price: {$lte: 400}})
```

Explanation: Finds books whose price is less than or equal to 400.

```
> db.books.find({price: {$lte: 400}})
< {
  _id: ObjectId('6ab2971a79f427617fd6afa4'),
  bookId: 6,
  bookName: 'The Alchemist',
  author: 'Paulo Coelho',
  price: 300,
  category: 'Fiction',
  quantity: 7
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa6'),
  bookId: 8,
  bookName: 'Ikigai',
  author: 'Hector Garcia',
  price: 350,
  category: 'Self Help',
  quantity: 11
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa8'),
  bookId: 10,
  bookName: 'Wings of Fire',
  author: 'A.P.J. Abdul Kalam',
  price: 250,
  category: 'Biography',
  quantity: 10
}
```

## 10. UPDATE OPERATION

MongoDB Query:

```
db.books.updateOne({bookId:5}, {$set:{price:600}})
```

Explanation: Updates the price of the book with bookId 5 to 600.

```
> db.books.updateOne({bookId:5}, {$set:{price:600}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

## 11. DELETE ONE OPERATION

MongoDB Query:

```
db.books.deleteOne({bookId:10})
```

Explanation: Deletes the book with bookId 10.

```
> db.books.deleteOne({bookId:10})
{
  acknowledged: true,
  deletedCount: 1
}
```

## 12. DELETE MANY OPERATION

MongoDB Query:

```
db.books.deleteMany({price: {$lt: 300}})
```

Explanation: Deletes all books whose price is less than 300.

```
> db.books.deleteMany({price: {$lt: 300}})
< {
  acknowledged: true,
  deletedCount: 2
}
```

### 13. AND OPERATION — \$and

MongoDB Query:

```
db.books.find({$and:[{price:{$gt:500}},{category:"Self Help"}]})
```

Explanation: Finds books with price greater than 500 and category Self Help.

```
> db.books.find({$and:[{price:{$gt:500}},{category:"Self Help"}]})
< {
  _id: ObjectId('6ab2971a79f427617fd6afa3'),
  bookId: 5,
  bookName: 'Atomic Habits',
  author: 'James Clear',
  price: 600,
  category: 'Self Help',
  quantity: 12
}
{
  _id: ObjectId('6ab2971a79f427617fd6afbc'),
  bookId: 30,
  bookName: 'The Power of Now',
  author: 'Eckhart Tolle',
}
```

## 14. OR OPERATION — \$or

MongoDB Query:

```
db.books.find({$or:[{category:"Fiction"},{category:"Fantasy"}]})
```

Explanation: Finds books whose category is Fiction or Fantasy.

```
db.books.find({$or:[{category:"Fiction"},{category:"Fantasy"}]})
{
  _id: ObjectId('6ab2971a79f427617fd6afa4'),
  bookId: 6,
  bookName: 'The Alchemist',
  author: 'Paulo Coelho',
  price: 300,
  category: 'Fiction',
  quantity: 7
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa7'),
  bookId: 9,
  bookName: 'Harry Potter',
  author: 'J.K. Rowling',
  price: 600,
  category: 'Fantasy',
  quantity: 6
}
{
  _id: ObjectId('6ab2971a79f427617fd6afac'),
  bookId: 14,
  bookName: 'The Hobbit',
  author: 'J.R.R. Tolkien',
  price: 550,
  category: 'Fantasy',
  quantity: 7
}
```

## 15. SORT OPERATION

MongoDB Query:

```
db.books.find().sort({price:1})
```

Explanation: Sorts books by price in ascending order.

```
> db.books.find().sort({price:1})
< {
  _id: ObjectId('6ab2971a79f427617fd6afa4'),
  bookId: 6,
  bookName: 'The Alchemist',
  author: 'Paulo Coelho',
  price: 300,
  category: 'Fiction',
  quantity: 7
}
{
  _id: ObjectId('6ab2971a79f427617fd6afab'),
  bookId: 13,
  bookName: 'Think and Grow Rich',
  author: 'Napoleon Hill',
  price: 300,
  category: 'Self Help',
  quantity: 9
}
{
  _id: ObjectId('6ab2971a79f427617fd6afb9'),
  bookId: 27,
  bookName: 'Sherlock Holmes',
  author: 'Arthur Conan Doyle',
  price: 300
}
```

**B. MongoDB Query:**

```
b.books.find().sort({price:-1})
```

Explanation: Sorts books by price in descending order.

```
> db.books.find().sort({price:-1})
< {
  _id: ObjectId('6ab2971a79f427617fd6afb6'),
  bookId: 24,
  bookName: 'Artificial Intelligence',
  author: 'Stuart Russell',
  price: 950,
  category: 'Technology',
  quantity: 5
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa2'),
  bookId: 4,
  bookName: 'Machine Learning',
  author: 'Tom Mitchell',
  price: 900,
  category: 'Technology',
  quantity: 5
}
{
  _id: ObjectId('6ab2971a79f427617fd6afb5'),
  bookId: 23,
  bookName: 'Database Systems',
  author: 'Raghu Ramakrishnan',
  price: 900,
  category: 'Technology',
  quantity: 4
}
```

## 16. LIMIT OPERATION

MongoDB Query:

```
db.books.find().limit(5)
```

Explanation: Displays only the first 5 books.

```
>_MONGOSH
> db.books.find().limit(5)
< {
  _id: ObjectId('6ab295f279f427617fd6af9f'),
  bookId: 1,
  bookName: 'Python Basics',
  author: 'Mark Lutz',
  price: 500,
  category: 'Technology',
  quantity: 10
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa0'),
  bookId: 2,
  bookName: 'Java Programming',
  author: 'Herbert Schildt',
  price: 700,
  category: 'Technology',
  quantity: 8
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa1'),
  bookId: 3,
  bookName: 'Data Science',
  author: 'John Smith',
  price: 800,
  category: 'Technology',
  quantity: 6
}
```

## 17. SEARCH BY CITY

MongoDB Query:

```
db.books.find({category:"Technology"})
```

Explanation: Finds all books belonging to the Technology category.

```
>_MONGOSH
>
> db.books.find({category:"Technology"})
< {
  _id: ObjectId('6ab295f279f427617fd6af9f'),
  bookId: 1,
  bookName: 'Python Basics',
  author: 'Mark Lutz',
  price: 500,
  category: 'Technology',
  quantity: 10
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa0'),
  bookId: 2,
  bookName: 'Java Programming',
  author: 'Herbert Schildt',
  price: 700,
  category: 'Technology',
  quantity: 8
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa1'),
  bookId: 3,
  bookName: 'Data Science',
  author: 'John Smith',
  price: 800,
  category: 'Technology',
  quantity: 6
}
```

## 18. COUNT DOCUMENTS

MongoDB Query:

```
db.books.countDocuments()
```

Explanation: Counts the total number of books in the collection.

```
> db.books.countDocuments()
```

```
< 47
```

```
library> |
```

## MONGODB OPERATORS USED:

| Operator     | Meaning                     | Example           |
|--------------|-----------------------------|-------------------|
| <b>\$eq</b>  | Equal to                    | marks: {\$eq:88}  |
| <b>\$gt</b>  | Greater than                | marks: {\$gt:85}  |
| <b>\$lt</b>  | Less than                   | marks: {\$lt:60}  |
| <b>\$gte</b> | Greater than or equal to    | marks: {\$gte:90} |
| <b>\$lte</b> | Less than or equal to       | marks: {\$lte:70} |
| <b>\$and</b> | All conditions true         | \$and:[...]       |
| <b>\$or</b>  | At least one condition true | \$or:[...]        |

## CONCLUSION

This project provided practical knowledge of NoSQL databases using MongoDB. A Student Management System containing student information was created and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update, and delete documents. I also learned how to Ascending order & descending order data, comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte work, along with logical operators such as \$and and \$or.

Overall, this project helped me understand how MongoDB can be used to manage and query student data efficiently.